

Indian Statistical Institute, Bangalore
B. Math I, First Semester, 2025-26
Final Examination
Probability Theory I
Maximum Score 100

10.11.25

Duration: 3 Hours

Calculators and two pages (one sided) of notes allowed.
Each question is worth 15 points.

1. In any ordered sequence of elements of two kinds, each maximal subsequence of elements of like kind is called a run. For example the sequence $\alpha\alpha\alpha\beta\alpha\alpha\beta\beta\beta\alpha$ starts with a run of length 3, followed by runs of length 1, 2, 3, 1, respectively. Given a indistinguishable α 's and b indistinguishable β 's
 - (a) How many distinguishable orderings are there?
 - (b) If there are n runs of α , what are the possible number of runs of β ?
 - (c) Given that all the distinguishable orderings are equally likely, what is the probability that there are n runs of α and $n - 1$ runs of β ?
2. Die A has four red and two white faces, whereas die B has three red and three white faces. A coin is flipped once. If it falls heads, the game continues by throwing die A alone; if it falls tails, die B is to be used.
 - (a) Find the probability of red at the first throw.
 - (b) If the first two throws resulted in white, what is the probability of red at the third throw?
 - (c) If white turns up at the first 3 throws, what is the probability that die B is being used?
3. One hundred independent rolls of a fair die will be made.
 - (a) Compute an approximation to the probability that the number 6 will appear between 12 and 22 times (inclusively).
 - (b) If number 6 appears exactly 25 times, find the approximate probability that number 5 will appear less than 20 times.
4. Suppose that the lifetime of an electronic light bulb has the hazard rate function

$$\lambda(t) = .25 + .14t^2, \quad t > 0,$$

where t is in years. What is the probability that

- (a) the light bulb survives to age 4 years?
 - (b) the light bulb's lifetime is between 2 and 4 years?
 - (c) a 2-year-old light bulb will survive to age 4 years?
5. Let $X \sim \text{Exp}(\lambda)$.
 - (a) The floor function $\lfloor x \rfloor = \max\{n \in \mathbb{Z} : n \leq x\}$ gives the largest integer less than or equal to x . Find the probability mass function of $\lfloor X \rfloor$.
 - (b) The fractional part of x is defined as $\{x\} = x - \lfloor x \rfloor$. Find the probability density function of $\{X\}$
6. A six-faced fair die is rolled repeatedly independently till 1 appears. Let X be the total number of rolls and Y be the number of times 6 appeared in these X rolls. So Y is the number of times 6 appears before the first 1 appears in repeated rollings of a die .
 - (a) Find $E[Y \mid X = x]$ and $\text{Var}[Y \mid X = x]$.
 - (b) Find $E(Y)$ and $\text{Var}(Y)$

7. Consider a pair (X, Y) of discrete random variables with joint PMF given by

		Y			
		1	2	3	4
X	0	0.1	0	0.1	0.1
	1	0.1	0.2	0.1	0.1
	2	0	0.1	0	0.1

- Find the marginal distributions of X and Y .
- Find the conditional distribution of X given $Y=3$.
- Are X and Y independent? Justify your answer.

[illegible]